

Consistent Partial Least Squares: A cSEM tutorial

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Abstract

This tutorial article provides a comprehensive, step-by-step illustration of consistent partial least squares (PLSc) using the **cSEM** package in R. PLSc is a variant of partial least squares path modeling, a composite-based approach to structural equation modeling, that produces consistent parameter estimates for latent variable models. In particular, PLSc corrects construct correlations for attenuation using Dijkstra-Henseler's composite reliability coefficient ϱ_A . Building on the employee retention example introduced by Cheah, Sarstedt, Hair, and Ringle (2026), we demonstrate how to install and load **cSEM**, specify the model using **lavaan**-style syntax, estimate the PLSc-SEM model, evaluate measurement model quality (internal consistency reliability, convergent validity, and discriminant validity), and assess the structural model (collinearity, effect sizes, model fit, and path coefficients with bootstrap inference). We further contrast the PLSc-SEM results with standard PLS-SEM estimates and discuss practical considerations for using PLSc-SEM in empirical research.

Keywords: measurement error, bias correction, partial least squares structural equation modeling (PLS-SEM), factor score regression

1 Introduction

Structural equation modeling (SEM) is a versatile multivariate analysis framework used in the behavioral, social, medical, and management sciences to study complex relationships between constructs and their relations with observed variables (e.g., Kline, 2023; Marcoulides & Schumacker, 1996). To estimate structural equation models, various approaches have been proposed, including maximum-likelihood (Jöreskog, 1970), weighted least squares (Browne, 1974), factor score regression (Devlieger & Rosseel, 2017; Skrandal & Laake, 2001) and (integrated) generalized structured component analysis (Hwang et al., 2021; Hwang & Takane, 2004)

A composite-based estimator to SEM that gained popularity over the last decades in various fields of social sciences (Cook & Forzani, 2023), including marketing (Sarstedt et al., 2022), information systems research (Benitez, Henseler, Castillo, & Schubert, 2020; Evermann & Rönkkö, 2023), human resource management (Ringle, Sarstedt, Mitchell, & Gudergan, 2020), and tourism research (Müller, Schubert, & Henseler, 2018), is partial least squares path modeling (PLS-PM, Wold, 1982), also known as partial least squares structural equation modeling (PLS-SEM, Hair, Ringle, & Sarstedt, 2011). PLS-PM is a two-step estimation procedure (Schubert, Zaza, & Henseler, 2023). In the first step, construct scores are obtained by the partial least squares algorithm. Subsequently, these construct scores are used to estimate the model parameters. Although PLS-PM is computationally efficient, it is known that in its original form to produce inconsistent estimates for models involving latent variables due to attenuation bias (e.g., Dijkstra, 1985; Hui & Wold, 1982; Rönkkö & Evermann, 2013; Schubert, Rosseel, et al., 2023).

To address this limitation, consistent partial least squares (PLSc, Dijkstra, 2013; Dijkstra & Henseler, 2015a, 2015b; Dijkstra & Schermelleh-Engel, 2014) was introduced. PLSc is based on PLS-PM, but applies a correction for attenuation to obtain consistent estimates of reflective measurement models and path coefficient between latent variables. Additionally, PLSc has been extended to deal with ordinal variables (Schubert, Henseler, & Dijkstra, 2018), randomly occurring outliers (Schamberger, Schubert, Henseler, & Dijkstra, 2020) and multicollinearity issues (Jung & Park, 2018).

To apply PLS-PM and PLSc, various implementations have been developed (Venturini, Mehmetoglu, & Latan, 2023). They can be distinguished into: (i) proprietary software and (ii) open-source software. Proprietary software for conducting PLS-PM and PLSc includes SmartPLS (Ringle, Wende, & Becker, 2024), WarpPLS (Kock, 2025) and ADANCO (Henseler, 2025). On the other hand, open-source software alternatives include implementations in the statistical programming environment R (R Core Team, 2025) such as *matrixpls* (Rönkkö, 2022), *SEM-inR* (Ray, Danks, & Calero Valdez, 2026), *cSEM* (Rademaker & Schubert, 2020) and *plspm* (Sanchez, Trinchera, Russolillo, & Bertrand, 2025). Similarly, it includes implementations in python (Python Software Foundation, 2024) such as *plspm* (Humble, 2024). While proprietary software such as SmartPLS often shows high ease of use (Cheah, Magno, & Cassia, 2024), they conflict with the open science principles (UNESCO, 2021). For example, their codebase is not openly accessible of transparency (Morin et al., 2012), reproducibility (Ince, Hatton, & Graham-Cumming, 2012), and accessibility/inclusiveness (). Open-source software overcomes these limitations (Barba, 2022). Therefore, this tutorial presents the R package *cSEM* as a full-fledged open-source alternative to SmartPLS to apply PLS-PM and PLSc.

2 cSEM R Illustration

2.1 Getting started with R

To use the cSEM package, users first need to install R and an integrated development environment (IDE) such as RStudio. For the installation, we refer to the official guides for R (<https://cran.r-project.org>) and RStudio (<https://posit.co/download/rstudio-desktop>).

When RStudio is opened for the first time, it displays four panes (Figure 1). The two most important ones are the *script editor* (upper left), where code is written and saved, and the *Console* (lower left), where the code is executed and results are printed. The *Environment* pane (upper right) lists the objects currently loaded into memory, and the fourth pane (lower right) provides access to files, plots, and help pages.

Although code can be typed directly into the console, it is good practice to work in a script, since a script can be saved, shared, and re-run, which is essential for reproducible analyses. To create a new script, click **File** → **New File** → **R Script**. The typical workflow is to write code in the editor and run the selected line(s) with **Ctrl + Enter**. Any line beginning with **#** is treated as a comment: it documents what the code does and is ignored when the code runs.

```
# This is a comment and is not executed.  
1 + 1 # Code is executed; the result is printed in the Console.  
  
[1] 2
```

Before cSEM can be used, it must be installed. The cSEM package is available on CRAN, so the most recent stable version can be installed with:

```
install.packages("cSEM")
```

Installation only needs to be done once per machine (rerun this line to update the package later). Moreover, development versions can be installed from GitHub: <https://github.com/FloSchuberth/cSEM>. After installation, the package must be loaded at the start of every new R session before its functions become available:

```
library(cSEM)
```

With cSEM installed and loaded, we can now import the data and specify the model, as described in the following sections.

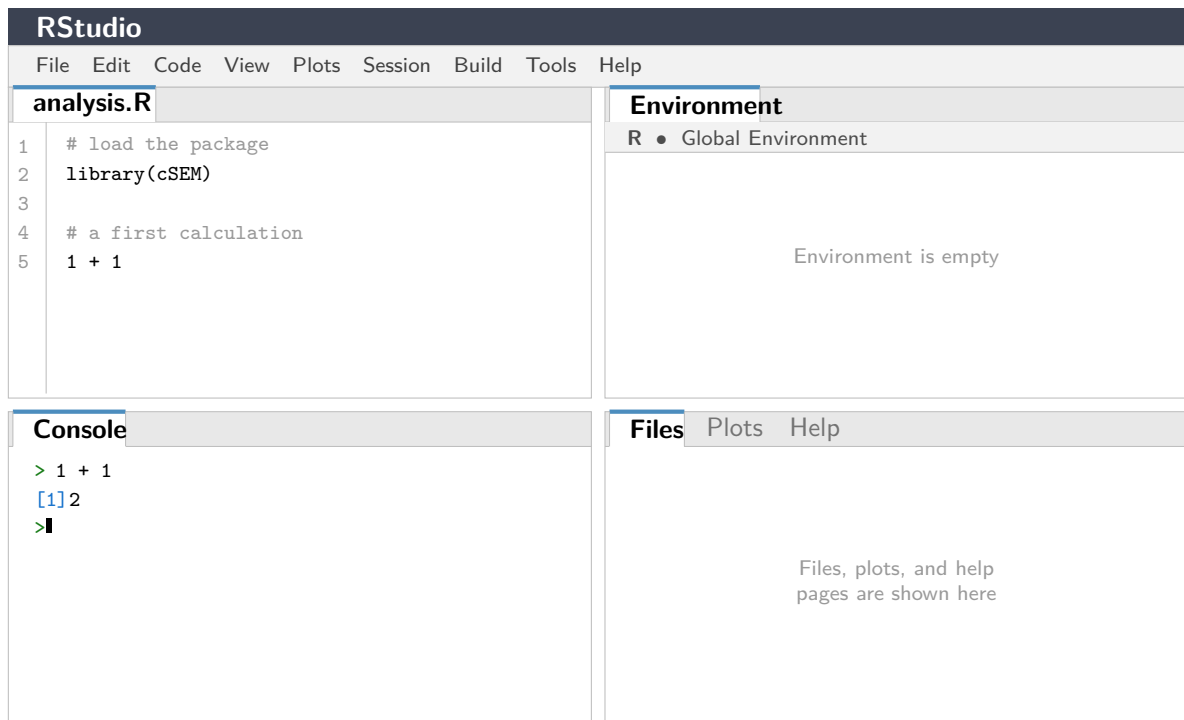


Figure 1: The four panes of the RStudio interface: the script editor (upper left), the *Console* (lower left), the *Environment* pane (upper right), and the files/plots/help pane (lower right).

2.2 Model and Data

The *cSEM* package (Rademaker & Schubert, 2020) offers a variety of composite-based approaches to SEM including PLS-PM and PLSc. Following Cheah et al. (2026), our illustration of the application of PLSc using *cSEM* draws on the employee retention model introduced by Hair, Black, Babin, and Anderson (2019, Chapter 13).

The model explains the effects of organizational commitment (OC) and job satisfaction (JS) on employees' staying intentions (SI), with work environment perceptions (EP) and attitudes toward coworkers (AC) as exogenous antecedent constructs. Five constructs are reflectively measured by a total of 21 indicators. A detailed description of the indicators can be found in Cheah et al. (2026). The model is presented in Figure 2.

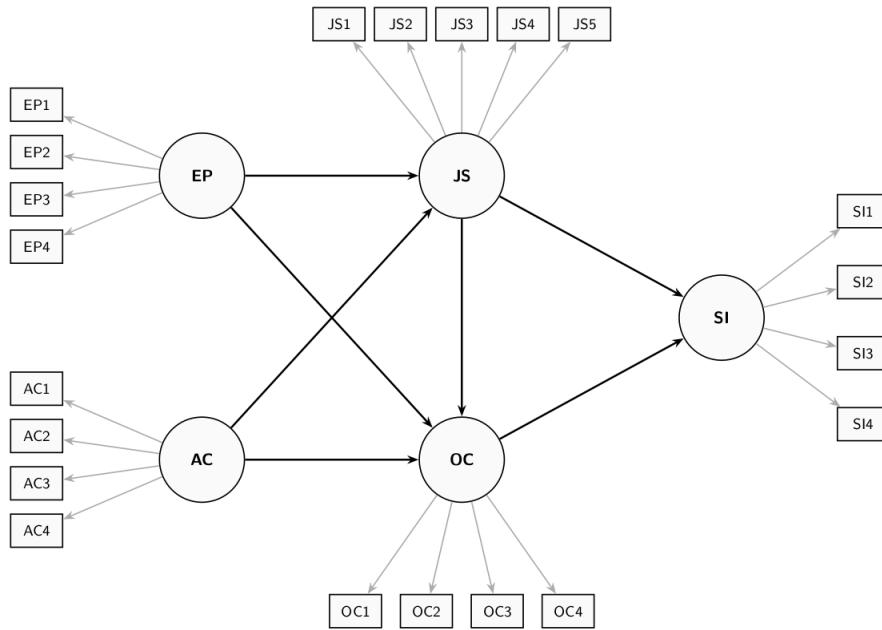


Figure 2: Employee Retention Model

The dataset used for model estimation is publicly available from the SmartPLS website: <https://www.smartpls.com/documentation/sample-projects/employee-retention>. The synthetic dataset comprises $N = 400$ observations from HBAT Industries and was generated to satisfy the assumptions of factor-based SEM, including multivariate normality. For this tutorial, we assume the data has been downloaded as a CSV file. After downloading the dataset, it needs to be loaded into the R Environment. The Employee Retention dataset is stored as an CSV File using ';' as the separator, thus it can be loaded into R using following command:

```
# Import dataset
HBAT_SEM <- read.csv("data/HBAT_SEM.csv", sep=";")

# Select relevant variables
HBAT_SEM_rel <- HBAT_SEM[, c("AC1", "AC2", "AC3", "AC4",
                             "EP1", "EP2", "EP3", "EP4",
                             "JS1", "JS2", "JS3", "JS4", "JS5",
                             "OC1", "OC2", "OC3", "OC4",
                             "SI1", "SI2", "SI3", "SI4")]

# show dimensions of the relevant dataset
dim(HBAT_SEM_rel)

[1] 400 21

# count missing values per variable
colSums(is.na(HBAT_SEM_rel))

AC1 AC2 AC3 AC4 EP1 EP2 EP3 EP4 JS1 JS2 JS3 JS4 JS5 OC1 OC2 OC3 OC4 SI1 SI2 SI3
0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0
SI4
0
```

Since `cSEM` can only handle complete datasets, i.e., datasets without any missing values, we check the number of missing values per variable using the `is.na()` function. The output shows

that the variables AC4 and EP4 each contain one missing values. Before we can continue, we thus need to either apply imputation methods or delete the incomplete observations. Deleting incomplete observations can be done via following command:

```
# filter the dataset, only select complete cases
HBAT_SEM_rel <- HBAT_SEM_rel[complete.cases(HBAT_SEM_rel),]

# the number of rows changed from 400 to 398
dim(HBAT_SEM_rel)

[1] 398  21
```

Since two observations contain missing values, all analyses in the remainder of this tutorial are based on the $N = 398$ complete observations. When using their own data, users must ensure it is stored as a data frame or a matrix before passing it to **cSEM** functions; if necessary, the data can be converted with `as.data.frame()` or `as.matrix()`.

2.3 Model specification

In **cSEM**, models are specified using the **lavaan** model syntax. The **lavaan** package does not need to be installed to use the syntax. In particular, the `=~` and the `<~` operators are used to define the relations between the constructs and their indicators. Specifically, the `=~` operator is used to specify reflective measurement models, while the `<~` is used to specify composite models (see , for an explanation of reflective and composite models). In addition, the `~` operator is used to specify the relations among constructs. In case, an observed variable should be directly accounted for in the structural model, it must be specified as a single indicator construct, as common in PLS-PM (). The **cSEM** specification of the retention model illustrated in Figure 2 is as follows:

```
model <- "
  # Measurement model
  AC =~ AC1 + AC2 + AC3 + AC4
  EP =~ EP1 + EP2 + EP3 + EP4
  JS =~ JS1 + JS2 + JS3 + JS4 + JS5
  OC =~ OC1 + OC2 + OC3 + OC4
  SI =~ SI1 + SI2 + SI3 + SI4

  # Structural model
  JS ~ EP + AC
  OC ~ EP + AC + JS
  SI ~ JS + OC
"
```

3 Estimating the model with cSEM

3.1 Running the estimation

The central function is `csem()`. The minimal input required for the `csem()` function is a dataset and a model:

.data A data frame or matrix with column names matching the indicator names used in the model description. Possible column types of data proved are: "logical", "numeric" ("double" or "integer"), "factor" ("ordered" and/or "unordered"), "character" or a mix of several types.

.model A model string written in *lavaan* syntax, with `=~` for measurement relations and `~` for structural (regression) relations.

By default weights are estimated using the partial least squares path modeling algorithm ("PLS-PM").

To ultimately estimate with the `csem` function, we call it and save the results as `res_csem`:

```
# PLSc-SEM estimation
res_csem <- csem(
  .data = HBAT_SEM_rel,
  .model = model
)
```

To obtain a comprehensive overview of all results, including loadings, path coefficients, reliability measures, and R^2 values, use:

```
# Full results summary
summary_res <- summarize(res_csem)
print(summary_res)
```

```
# For a more compact overview
summary_res$Estimates>Loading_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	EP =~ EP1	Common factor	0.6341011	NA	NA	NA
2	EP =~ EP2	Common factor	0.8786819	NA	NA	NA
3	EP =~ EP3	Common factor	0.7678021	NA	NA	NA
4	EP =~ EP4	Common factor	0.8230691	NA	NA	NA
5	AC =~ AC1	Common factor	0.7691299	NA	NA	NA
6	AC =~ AC2	Common factor	0.6765259	NA	NA	NA
7	AC =~ AC3	Common factor	0.8215860	NA	NA	NA
8	AC =~ AC4	Common factor	0.9933858	NA	NA	NA
9	JS =~ JS1	Common factor	0.6248337	NA	NA	NA
10	JS =~ JS2	Common factor	0.6964587	NA	NA	NA
11	JS =~ JS3	Common factor	0.7182274	NA	NA	NA
12	JS =~ JS4	Common factor	0.6319045	NA	NA	NA
13	JS =~ JS5	Common factor	0.9004003	NA	NA	NA
14	OC =~ OC1	Common factor	0.4252105	NA	NA	NA
15	OC =~ OC2	Common factor	0.9574682	NA	NA	NA
16	OC =~ OC3	Common factor	0.6858765	NA	NA	NA
17	OC =~ OC4	Common factor	0.8565670	NA	NA	NA
18	SI =~ SI1	Common factor	0.8170099	NA	NA	NA
19	SI =~ SI2	Common factor	0.8705194	NA	NA	NA
20	SI =~ SI3	Common factor	0.6778349	NA	NA	NA
21	SI =~ SI4	Common factor	0.8959341	NA	NA	NA

```
summary_res$Estimates$Path_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	JS ~ EP	Common factor	0.244824960	NA	NA	NA
2	JS ~ AC	Common factor	-0.003282073	NA	NA	NA
3	OC ~ EP	Common factor	0.428781485	NA	NA	NA
4	OC ~ AC	Common factor	0.172554413	NA	NA	NA
5	OC ~ JS	Common factor	0.096069429	NA	NA	NA
6	SI ~ JS	Common factor	0.131226588	NA	NA	NA
7	SI ~ OC	Common factor	0.490013776	NA	NA	NA

Note that the standard errors, t -values, and p -values in the `summarize()` output are reported as NA. The reason is that a plain `csem()` call computes point estimates only. To conduct statistical inference, the standard errors of the estimates must be obtained by a resampling procedure such as the bootstrap, which is introduced in Section 4.

4 Assessment methods in cSEM

The next step involves assessing the model estimates, according to the guidelines described in Benitez et al. (2020). Before any results are interpreted, we first validate the estimation. Second we assess the model adequacy, with $SRMR$, d_{uls} and d_G . Third we subsequently examine the reliability of construct scores, with the Dijkstra–Henseler’s ρ_A . Fourth we check the factor loading estimates and significance. Fifth we assess construct validity, i.e. convergent and discriminant validity (Campbell & Fiske, 1959). Convergent validity is assessed via the average variance extracted AVE . Discriminant validity is assessed with the $HTMT$ and its improved version $HTMT2$. Sixth we check for Multicollinearity, via VIF . Seventh we check the weights and loadings values and significance.

Validation of the estimation

```
# Check whether the estimation is admissible
```

```
verify(res_csem)
```

```
-----
```

```
Verify admissibility:
```

```
admissible
```

```
Details:
```

Code	Status	Description
1	ok	Convergence achieved
2	ok	All absolute standardized loading estimates <= 1
3	ok	Construct VCV is positive semi-definite
4	ok	All reliability estimates <= 1
5	ok	Model-implied indicator VCV is positive semi-definite

```
-----
```

Testing the adequacy of the model

```
# Bootstrap-based test of the overall model fit
fit_test <- testOMF(
  res_csem,
  .R = 1000,
  #.seed = 42,
  .handle_inadmissibles = "replace"
  #.Random.seed = 42
)
```

```
Error in 'testOMF()':
! object '.Random.seed' not found
```

```
fit_test
```

```
Error:
! object 'fit_test' not found
```

Evaluating the reliability of the construct scores

The `assess()` function computes all relevant quality criteria:

```
#Compute all quality criteria
quality <- assess(res_csem)
```

```
Warning: The following warning occurred in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning: The following warning occurred in the calculateHTMT() function:
Intra-block and inter-block correlations between indicators must be either all-positive
or all-negative.
Warning in log(x[[3]]): NaNs produced
```

```
# Internal consistency reliability
quality$Reliability

$Cronbachs_alpha
      EP      AC      JS      OC      SI
0.8595716 0.8936315 0.8450310 0.8328395 0.8890480

$Joereskogs_rho
      EP      AC      JS      OC      SI
0.8607098 0.8918737 0.8417418 0.8343858 0.8901518

$`Dijkstra-Henselers_rho_A`
      EP      AC      JS      OC      SI
0.8717254 0.9105299 0.8568177 0.8878870 0.8989122
```

The results indicate that all constructs meet the commonly accepted thresholds. All constructs succeed the recommend threshold of 0.707 for Dijkstra-Henseler ρ_a , indicating reliable construct scores. Additionnaly `assess()` outputs alternative reliability estimates.

Evaluating indicator reliability

```
# Re-estimate with bootstrap resampling for inference
res_csem_boot <- csem(
  .data          = HBAT_SEM_rel,
  .model         = model,
  .resample_method = "bootstrap",
  .R             = 1000,
  .seed         = 42
)
```

Warning: package 'future' was built under R version 4.5.3

```
# Loading estimates incl. bootstrap std. errors, t-values and p-values
summarize(res_csem_boot)$Estimates$Loading_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	EP =~ EP1	Common factor	0.6341011	0.08825730	7.184687	6.736133e-13
2	EP =~ EP2	Common factor	0.8786819	0.06817644	12.888350	5.235172e-38
3	EP =~ EP3	Common factor	0.7678021	0.08865615	8.660449	4.699094e-18
4	EP =~ EP4	Common factor	0.8230691	0.06234019	13.202864	8.446117e-40
5	AC =~ AC1	Common factor	0.7691299	0.08208943	9.369415	7.293568e-21
6	AC =~ AC2	Common factor	0.6765259	0.09899325	6.834061	8.254389e-12
7	AC =~ AC3	Common factor	0.8215860	0.07771581	10.571671	4.032442e-26
8	AC =~ AC4	Common factor	0.9933858	0.04413543	22.507672	3.491295e-112
9	JS =~ JS1	Common factor	0.6248337	0.11846811	5.274278	1.332798e-07
10	JS =~ JS2	Common factor	0.6964587	0.10384121	6.706959	1.987220e-11
11	JS =~ JS3	Common factor	0.7182274	0.10374878	6.922755	4.429422e-12
12	JS =~ JS4	Common factor	0.6319045	0.12166320	5.193883	2.059521e-07
13	JS =~ JS5	Common factor	0.9004003	0.09311164	9.670115	4.039225e-22
14	OC =~ OC1	Common factor	0.4252105	0.06655371	6.388983	1.669924e-10
15	OC =~ OC2	Common factor	0.9574682	0.03086106	31.025128	2.470773e-211
16	OC =~ OC3	Common factor	0.6858765	0.05274373	13.003943	1.161946e-38
17	OC =~ OC4	Common factor	0.8565670	0.04700870	18.221458	3.487351e-74
18	SI =~ SI1	Common factor	0.8170099	0.04630598	17.643722	1.137185e-69
19	SI =~ SI2	Common factor	0.8705194	0.03913952	22.241445	1.364890e-109
20	SI =~ SI3	Common factor	0.6778349	0.05957767	11.377332	5.423357e-30
21	SI =~ SI4	Common factor	0.8959341	0.04423967	20.251826	3.423134e-91
	CI_percentile.95%L	CI_percentile.95%U				
1	0.4532620	0.7798607				
2	0.7214431	0.9805613				
3	0.5959517	0.9257479				
4	0.6938317	0.9348749				
5	0.6214292	0.9414872				
6	0.4665006	0.8574168				
7	0.6735746	0.9734165				
8	0.8318356	0.9978215				
9	0.3740575	0.8383272				
10	0.4843479	0.8895589				
11	0.4989930	0.8887790				
12	0.3733224	0.8466869				
13	0.6369939	0.9889399				

14	0.2851689	0.5408111
15	0.8851819	0.9963862
16	0.5873668	0.7951513
17	0.7585917	0.9460804
18	0.7268224	0.9095172
19	0.7801786	0.9365719
20	0.5648917	0.7977440
21	0.8073558	0.9718306

Evaluating convergent validity

```
# Average variance extracted (AVE)
quality$AVE
```

	EP	AC	JS	OC	SI
	0.6102822	0.6777668	0.5202693	0.5754208	0.6718668

The average variance extracted (AVE) for all constructs are above the recommended threshold of 0.5, supporting convergent validity.

Evaluating discriminant validity

```
# Heterotrait-monotrait ratio of correlations (HTMT)
quality$HTMT
```

```
$htmts
      EP      AC      JS      OC SI
EP 1.0000000 0.0000000 0.0000000 0.0000000 0
AC 0.2555939 1.0000000 0.0000000 0.0000000 0
JS 0.2437979 0.05239146 1.0000000 0.0000000 0
OC 0.4939933 0.27449892 0.2086788 1.0000000 0
SI 0.5672796 0.30936763 0.2313363 0.5005239 1
```

```
$quantiles
NULL
```

```
$nr_admissibles
NULL
```

Assessing multicollinearity among the indicators

```
# VIF values for Mode B (composite) weights
quality$VIF_modeB
```

Evaluating the weights

```
# Weight estimates incl. bootstrap std. errors, t-values and p-values
summarize(res_csem_boot)$Estimates$Weight_estimates
```

	Name	Construct_type	Estimate	Std_err	t_stat	p_value
1	EP <~ EP1	Common factor	0.2425256	0.03033059	7.996073	1.284500e-15
2	EP <~ EP2	Common factor	0.3360708	0.03417091	9.834998	7.956905e-23
3	EP <~ EP3	Common factor	0.2936625	0.02620371	11.206907	3.771439e-29
4	EP <~ EP4	Common factor	0.3148005	0.02512880	12.527478	5.281529e-36
5	AC <~ AC1	Common factor	0.2707114	0.02974091	9.102324	8.841910e-20
6	AC <~ AC2	Common factor	0.2381175	0.03264047	7.295162	2.982994e-13
7	AC <~ AC3	Common factor	0.2891744	0.02794699	10.347249	4.306835e-25
8	AC <~ AC4	Common factor	0.3496430	0.01889744	18.502136	1.984503e-76
9	JS <~ JS1	Common factor	0.2223363	0.04059423	5.477043	4.324929e-08
10	JS <~ JS2	Common factor	0.2478229	0.03710726	6.678555	2.413101e-11
11	JS <~ JS3	Common factor	0.2555689	0.03888206	6.572926	4.933583e-11
12	JS <~ JS4	Common factor	0.2248524	0.04381154	5.132264	2.862779e-07
13	JS <~ JS5	Common factor	0.3203920	0.03598888	8.902527	5.458932e-19
14	OC <~ OC1	Common factor	0.1740754	0.02457108	7.084563	1.394839e-12
15	OC <~ OC2	Common factor	0.3919744	0.01869390	20.968041	1.284457e-97
16	OC <~ OC3	Common factor	0.2807885	0.02061076	13.623392	2.907433e-42
17	OC <~ OC4	Common factor	0.3506668	0.01644880	21.318690	7.616096e-101
18	SI <~ SI1	Common factor	0.2882324	0.01534997	18.777390	1.156254e-78
19	SI <~ SI2	Common factor	0.3071099	0.01550302	19.809683	2.456267e-87
20	SI <~ SI3	Common factor	0.2391329	0.01876727	12.742019	3.453427e-37
21	SI <~ SI4	Common factor	0.3160760	0.01634721	19.335168	2.717827e-83
	CI_percentile.95%L		CI_percentile.95%U			
1	0.1828087		0.2999783			
2	0.2725098		0.4008700			
3	0.2414227		0.3459511			
4	0.2697347		0.3683087			
5	0.2183269		0.3399151			
6	0.1723162		0.2988541			
7	0.2386440		0.3504807			
8	0.2919151		0.3651913			
9	0.1423157		0.3013401			
10	0.1794724		0.3181615			
11	0.1792914		0.3324710			
12	0.1395502		0.3054149			
13	0.2341224		0.3685690			
14	0.1229501		0.2179109			
15	0.3530282		0.4250516			
16	0.2467718		0.3260126			
17	0.3185797		0.3842689			
18	0.2590976		0.3185720			
19	0.2763323		0.3387036			
20	0.2055149		0.2781864			
21	0.2820537		0.3429069			

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